Progressive exercises

E1 · Practice. Calculate $\displaystyle\int_0^2 (3x^2 - 2x + 1)\ \mathrm{d}x$ by finding an antiderivative.

E2 · Practice. In an orthonormal frame, calculate the area of the region bounded by the curves $y = x$ and $y = x^2$, between the abscissae 0 and 1. Give the result in area units.

E3 · Practice. Calculate the mean value of the exponential function on the interval $[0, 2]$.

E4 · Practice. Using integration by parts, calculate $\displaystyle\int_1^e \ln x\ \mathrm{d}x$.

E5 · Practice. For every $n \in \mathbb{N}$, let $I_n = \displaystyle\int_0^1 \frac{x^n}{1+x}\ \mathrm{d}x$, with $x^0 = 1$ in the integrand. Prove that $0 \leq I_n \leq \frac{1}{n+1}$, then deduce the limit of (I_n).

Textbook or past-paper work

Barbazo, p. 260, exercises 28 and 29. In S66: exercise 28, stating explicitly the two functions chosen for the integration by parts. At home: exercise 29, to adapt this choice to a product containing a logarithm.